

BlastSmart App

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1 Abstract

The blasting and mining industry rely on manual calculations, printed design guides, and paper-based reference materials for blast planning and design. These old methods can be time-consuming, prone to human error, and often inefficient in environments that require rapid engineering decision-making. This project presents the research, design, and development of BlastSmart Application, a mobile application developed to support blasting and explosives engineers in performing real-time blast design calculations and validating blasting parameters for both surface and underground mining operations.

2 Background

The application focuses on the explosives engineering curriculum to help the explosives engineers perform their job at a faster rate. The explosives engineers work with different formulas and must make a lot of assumptions to calculate what is needed for the blast holes. The have to consider a lot of factors such as environment working on, the type of rock (hard rock, medium rock and soft rock), what minerals exists in that rock, what kind of disturbances can happen during the operation, is it underground or the surface working on and others. It is difficult to remember all the information at once. Blast Smart App is created to simplify the work of engineers and provide them with trusted results.

3 Literature Review

Traditional pen and paper rock blasting procedures continuously prove to be inefficient for surface blast design and rock fragmentation processes. The implementation of the BlastSmart app provides a solution to these problems by digitalizing mining operations, ensuring less human induced errors that have continuously affected mine operations in the past. According to Morin and Ficarazzo (2006), rock fragmentation is considered the most important aspect of production blasting because it direectly affects the costs of drilling and blasting, loading, hauling and crushing. The BlastSmart app is intended to help minimize the occurences of flyrock, ground vibrations and poor fragmentation due to it's digitalized nature, thus ensuring efficient procedures for the Dwala Global Resources. Design parameters for the surface blast and fragmentation calculators include the burden(B), in-hole density(p), spacing to burden ratio(a), bench height(BH) and relative effective energy of the explosives(E). These parameters are meant to be used as inputs in order to calculate the stemming length(SL), Burden(B), actual powder factor(PFact) and the mean fragmentation size as per the following equations:

$$\mathrm{B} = \sqrt{\frac{M(BH-SL)}{a \cdot PF_{tech} \cdot BH}}$$

$$M = \frac{PD^2}{1273}$$

$PF_{act} = \frac{M(BH-SL\pm SD)}{B \cdot S \cdot BH}$

The sub-drill is given as a quarter or a third of the burden. For the Kuz-ram mean fragmentation model, the app uses cunningham's kuznetsov adapted equation to calculate the mean fragmentation size, which is given by the following equation :
$X_{50} = A\,k^{-0.8} Q^{0.167} \left(\frac{115}{E}\right)^{0.633}$

4 Design

4.1 Design Objective

BlastSmart App is to provide a tool to enable for the digitisation of traditionally paper intensive processes and integrate surface blasting design and fragmentation size calculation into an efficient tool. The blasting component of the calculator is supposed to input specific design parameters and give outputs. The fragmentation component is supposed to input variables defined by the Kuz-ram fragmentation model.

4.2 System logic

The code assigns blast geometry parameters, defining the data-types the user can input. Using these preliminary parameters, calculations regarding the explosive column can be obtained. The code is systematic, there is a break to allow the user to input more information needed to commence with execution. Conditional statements are used to restrict the inputs to within the range given in academia, so as to not overstate the capabilities of the code. Therefore, if the statement is true the block of code under the conditional statement executes. Otherwise, the next conditional statement and its respective block of code executes until all the assumptions are tested. If a conditional statement is found to be false the code only flags discrepancies.

4.3 Design Assumptions

stemming factor of 25 hole diameters was taken,as it provides a reasonable approximation for most blasting situations **sub-drill** taken as 1/3 to simplify calculations as brief does not call for subdrill calculation. **coupling percentage** taken as 100 percent since the brief did not call for coupling calculations.

5 Results and Discussion

5.1 Results

The algorithm, pseudo code, and python code give us the expected results with the assumptions we made. The calculators give us the values we need to give the explosives engineers. The first calculator gives the stemming length, the burden, and the actual powder factor used in the blast holes provided by other values to calculate them. We made some assumptions based on our target audience who already have knowledge about what the application is supposed to provide. Then the second calculator used some values calculated from the first one to produce the mean fragmentation size results using the Kuz-Ram Fragmentation Model based on the explosive curriculum. We combined the two calculators to the same algorithm and gave us the correct and accurate pseudo code and the python code.

5.2 Discussion of Results

The BlastSmart App performs surface blasting calculations and calculations predicting mean fragmentation size using the Kuz Ram model. The program runs in two stages: first, it calculates stemming length, burden, spacing, subdrill, and the actual powder factor from user inputs; second, it recalls these values and combines them with explosive energy and rock factor to estimate mean fragment size. Conditional checks ensure inputs fall within academic ranges, preventing the code from executing values that can skew results. The system validates first, keeping outputs consistent with empirical equation.

6 Conclusion

The BlastSmart app successfully transforms traditional pen and paper or manual design procedures into fully digitalized operations. The app guarantees a more enhanced and efficient way of performing mining operations calculations, and by so doing, the app helps cut major costs for Dwala Global Resources by optimizing time consuming processes and reducing human calculation errors. The software correctly computes burden, stemming, actual powder factor and the predicted mean fragmentation sizes.

7 Recommendations

For future development, BlastSmart will focus on expanding its engineering capabilities by incorporating advanced analytical tools such as predictive blast modelling, fragmentation analysis, ground vibration prediction, fly rock estimation, and air overpressure forecasting. These improvements will enable engineers to assess blast performance before implementation, leading to improved safety, operational efficiency, and environmental compliance.

8 References

References

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